

Nutworth: Designing a Context-Aware Spending Companion to Promote Value-Aligned Financial Decisions

Yee Kit Chan*
School of Computer Science
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
yeekitc@andrew.cmu.edu

Grace Malinger*
Department of Mechanical
Engineering
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
gmalinge@andrew.cmu.edu

Evelynn Chen*
Information Systems
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
evelynnnc@andrew.cmu.edu

Minjung Park†
Human-Computer Interaction
Institute
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
mpark2@andrew.cmu.edu

Joanna Ni*
School of Design
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
wenjunn@andrew.cmu.edu

Geoff Kaufman†
Human-Computer Interaction
Institute
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
gfk@cs.cmu.edu

Abstract

Young adults often struggle with everyday financial decision-making due to emotional stress, cognitive load, and impulsive spending rather than deliberate budgeting. As financial well-being becomes an increasing concern, we designed Nutworth, a context-aware spending companion that supports young adults to make more intentional financial decisions by translating abstract financial trade-offs into concrete metaphors and personalized interactions that encourage reflection. Through three iterative prototyping rounds, we gathered think-aloud and reflective feedback with 15 participants. We found that emotional safety, clear reflection-derived distinctions between spending goals, and autonomy-supportive framing are essential for reducing reactance and fostering engagement with long-term values through everyday choices. We discuss implications for persuasive financial technologies and propose open research questions for future study of long-term impacts on financial self-efficacy.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

Value-aligned spending, Financial decision-making, Persuasive design, Dual-process theories, Cognitive dissonance, Reflective design

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1 Introduction and Background

Many struggle to resist making impulsive, emotionally-driven financial decisions. Experienced stock traders make critical investment decisions based on cognitive and emotional biases [18] and college students often struggle to align daily financial decisions with long-term goals, frequently engaging in impulsive spending despite their intentions to save. Across these contexts, the gap between intention and action is largely emotional and context-dependent, driven by immediate, potentially irrational desires rather than deliberate reasoning [10, 16, 18]. Young adults are particularly susceptible to this challenge, as financial stress is prevalent among them and is associated with increased anxiety, lower financial self-efficacy, and poorer financial behaviors [11, 23]. They also lack foundational financial literacy, which undermines their sense of control and contributes to ineffective budgeting and saving practices [2]. Together, these factors contribute to feelings of overwhelm, guilt, and discouragement, reinforcing the disconnect between students' long-term values and everyday spending behavior.

To address these challenges, a range of financial tools and design approaches have emerged. Budgeting applications promote awareness through categorization, numerical visualization, and automated alerts [20], while automated savings tools support long-term accumulation [22]. More recent design and marketing research explores how financial technologies encourage healthier financial behaviors through gamification [1]. This work builds on extensive research in consumer behavior, sociology, and behavioral finance showing that financial decisions are shaped by emotions, social norms, and cognitive biases rather than purely rational calculations. [3, 6, 10, 16, 20, 20, 22].

Some financial tools operationalize these insights through theory-informed design. For example, dual process theory distinguishes how human cognition operates through two distinct processing modes: *System 1* is fast, automatic, and prone to bias, and *System 2* is slow and effortful, supporting complex reasoning and requiring

*First authors contributed equally to this research.

†Senior authors provided guidance and mentorship throughout this work.



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cognitive resources [10]. When making spending decisions, people often rely on System 1 processing, resulting in impulsive purchases that do not reflect long-term goals. PocketGuard's "leftover" feature applies this framework by computing a single, intuitive number from users' past and recurring purchase data and current goals, reducing the need for effortful System 2 thinking and empowering System 1 to make quick and responsible decisions without cognitive fatigue [17]. Similarly, *self-affirmation theory* suggests that reflecting on the perception of oneself provides a psychological buffer and supports long-term motivation [19]. For instance, Chase Credit Journey celebrates small credit score gains [9]. This provides a psychological buffer through visible progress and reinforces a positive perception of oneself as someone who manages credit, extending confidence to managing bank accounts and credit cards. This connects to *hypocrisy induction* [21] which can encourage reflection without imposing guilt by surfacing gaps between goals and actions. The Chase app operationalizes this through category budgets and a gentle overspending indicator [8].

The *broaden-and-build theory* demonstrates that positive emotions expand attention and cognitive flexibility [5], suggesting that supportive, nonjudgmental designs and visuals may sustain engagement better than guilt-based approaches. Chase again exemplifies this when users fall behind on long-term goals: rather than highlighting failure, the app offers friendly pop-ups acknowledging that balancing financial goals can be hard and offering advisor support [8]. Finally, *cognitive reframing* involves interpreting situations through alternative perspectives that shift their emotional impact [13]. Rocket Money's 'Safe-to-Spend' feature applies this by displaying actual spendable money after obligations rather than total account balance, helping users make more confident financial decisions [4].

Despite the growing recognition of emotional and cognitive dimensions in financial decision-making, existing tools exhibit significant limitations. Traditional services like Mint [7], Rocket Money [4], and PocketGuard [17] categorize expenditures and notify users when spending exceeds limits, but may overwhelm students with granular data while relying on negative reinforcement, adding to the embarrassment, shame, and anxiety financially-stressed students already face [12]. Automated savings tools such as Oportun [14] facilitate passive accumulation, but do not intervene at the moment spending decisions occur, overlooking opportunities for intervention and reflection at the moment of purchase. Although gamification through streaks and rewards can enhance motivation, it can also introduce pressure, comparison, or unintended identity effects [1, 24]. Furthermore, these services issue reminders on fixed schedules instead of at contextually relevant moments, which limits their effectiveness in interrupting impulsive spending.

A pattern emerges in retail banking, retirement planning, and everyday budgeting tools: while some incorporate elements of theory-informed design, they lack holistic and emotionally supportive experiences. Financial management is still largely treated as a rational process, despite research showing that spending is emotional and context-dependent. As a result, few tools meaningfully intervene at the moment of spending or support emotionally safe reflection on financial trade-offs.

Nutworth addresses this gap as a context-aware companion that prompts users to pause, reflect, and reconnect with their financial

goals through human-centered guidance rather than strict rules for everyday spending behavior. The system leverages signals such as location, merchant category, temporal patterns, checkout progress, and users' historical spending behaviors (e.g., frequency of visits, prior purchases at specific merchants or times) to provide personalized, situationally relevant interventions.

2 Design Process

We designed Nutworth in three iterative rounds of prototyping (Figure 1), each with five college-aged participants. We prioritized psychological aspects of financial decision-making over usability or technology feasibility with a *Wizard of Oz* approach. Our goal was to explore how design interventions grounded in cognitive psychology theories and metaphors can promote intentional financial reflection without inducing guilt, shame, or threat to users' self-concept. In each round, we aimed to (1) understand how tone, metaphors, and interventions impacted users and (2) reflect interactions based on qualitative feedback.

Participants completed structured, scenario-based think-aloud sessions with a Figma prototype of Nutworth, verbalizing interpretations, emotions, and decision-making processes while interacting with onboarding, trade-offs, and reflection features. We reviewed session notes to identify recurring patterns in confusion, emotional reactions, and perceptions of autonomy, and translated these to design refinements throughout the rounds.

2.1 Round 1

We aimed to evaluate users' initial interpretations of the onboarding flow, stash and goal concepts, and barter interaction, paying particular attention to whether reflection prompts and trade-off metaphors were clear, meaningful, and non-judgmental. The participants highlighted three key design implications to improve clarity, motivation, and emotional safety within the system. First, participants' spending patterns frequently deviated from their articulated financial values, suggesting the need for onboarding that prompted deliberate reflection. Second, many users initially assumed that all stashes represented long-term savings goals, indicating a strong need to clearly distinguish goals from *stashes*, which are weekly recurring resources designated for categorical spending. Finally, users often misunderstood how *trade-offs* functioned and described the interface as guilt-inducing.

To address this feedback, we refined the interactive onboarding experience and reframed trade-offs as informed choices. We changed the onboarding to guide users through articulating their financial values and setting up both goals and stashes. This encouraged users to engage in self-reflection by leveraging internally-generated inconsistency rather than succumbing to external judgment [21]. We also introduced the distinction between goals stashes early in the experience, allowing users to surface their historical progress and helping leverage principles of consistency and self-affirmation to strengthen engagement. Moreover, when presenting trade-offs, we framed them as informed choices rather than losses, reducing perceived moral judgment while preserving opportunities for reflection.

Furthermore, participants expressed mixed reactions to Nutworth's use of contextual data for habit-based nudges. While some

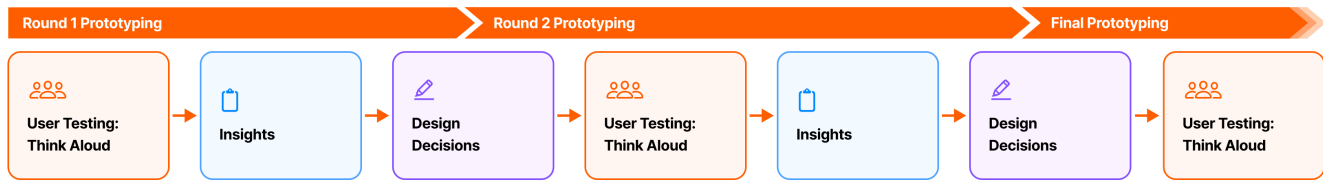


Figure 1: Overview of the iterative design process over three rounds.

found location and transaction-based personalization helpful, others viewed continuous tracking as invasive or unnecessary, or even “condescending”. To address these concerns, we introduced granular permission toggles, customizable notification settings, and a manual transaction input option. These changes allowed users to retain control over sensitive data while still benefiting from system insight.

2.2 Round 2

Building on Round 1 insights, we streamlined onboarding to support reflective creation of goals and stashes through visualizations. We aimed to clarify the concept of excess spending as trade-offs between stashes, utilizing ‘jar of acorns’ visuals for goals and stashes to mitigate negative affects. We tried to replace previous loss-framed visuals that made users feel judged or distressed. Purchases were not framed as good or bad, but as choices between valuable options through cognitive reframing, allowing users to compare between stashes or goals based on their own spending history.

We aimed to assess how the revised trade-off thresholds, reframing mechanisms, and weekly reflection prompts influenced users’ understanding of trade-offs, perceptions of autonomy, and anticipated engagement. This second round of user testing further revealed that users interpreted the squirrel companion in divergent ways, such as a companion, themselves, a mascot, or a strict monitor. This indicates a need to clarify representations as an accountability buddy. Participants also felt discouraged when stashes were completely depleted, often perceiving these moments as “already too late,” suggesting the importance of introducing threshold-based cues or interventions that emphasized manageability over negative reinforcement. Users indicated a greater willingness to engage with reflection when interactions felt lightweight, optional, or game-like, supporting brief, low-stakes prompts. These refinements increased the psychological distance from impulsive decisions, shifting user focus from short-term affordability to long-term value.

3 Design of Nutworth

Based on previous iterations, we designed Nutworth (Figure 2) as a context-aware, reflective spending companion that supports intentional financial decision-making without inducing guilt or anxiety. Rather than enforcing budgets or labeling spending as a success or failure, Nutworth frames spending as a series of meaningful trade-offs grounded in user-defined priorities. All screenshots can be found in the supplementary materials. Nutworth seamlessly implements persuasive design theories and additional features described in the following.

Nutworth’s design prioritizes emotional safety and self-consistency by explicitly drawing on self-affirmation theory, hypocrisy induction, and the broaden-and-build theory. During onboarding, users are prompted to articulate personal financial values, which functions as a self-affirmation intervention that buffers against identity threat during later feedback. Because nudges are anchored in user-authored priorities, moments of misalignment surface as internally generated inconsistency rather than external judgment, inviting hypocrisy induction and dissonance resolution while minimizing reactance. Throughout the interface, warm visual language, supportive tone, celebratory micro-interactions, and non-evaluative insights reflect principles of the broaden-and-build theory, leveraging positive affect to expand users’ openness to reflection and sustain engagement during potentially stressful financial moments.

At the moment of spending, Nutworth supports reflective decision-making while minimizing cognitive burden by leveraging dual process theory and cognitive reframing. Trade-off prompts presenting intuitive equivalencies and alternatives appear only when a potential purchase meaningfully impacts a stash or goal, such as when exceeding the weekly allotment and thus reducing available resources for other stashes. By framing purchases as choices between personally meaningful options rather than prohibitions or losses, Nutworth encourages deliberation and autonomy.

As an example case (Figure 3), the system identifies that a potential purchase would significantly impact the “Food” stash. The trade-off card displays the amount, contextualized through a playful equivalency: “That is equal to 4 Chipotle orders.” This metaphor leverages anchoring and relatability, helping users intuitively grasp the magnitude of the choice without numerical overload. The users are then offered two options: “Proceed Purchase” or “I will hold on to it.” Importantly, both options are framed as acceptable. To prioritize flexibility and user-defined meaning, a dropdown allows users to change which stash the purchase would draw from.

In some situations, the system reframes the trade-off in relation to a long-term goal rather than a stash. When the app predicts that redirecting the money towards a goal could yield significant progress, for example, “This could increase your progress in camera savings by 15%”, the card highlights this opportunity. This frame operationalizes the principle of value-congruent nudging, showing users not what they “should” do, but how their choice can more closely align with their previously stated aspirations. Again, users may proceed with the purchase or choose to save toward the goal. Selecting the savings option produces a celebratory screen featuring the squirrel that affirms the user’s action and updating the goal progress bar in real time. This moment of positive reinforcement is

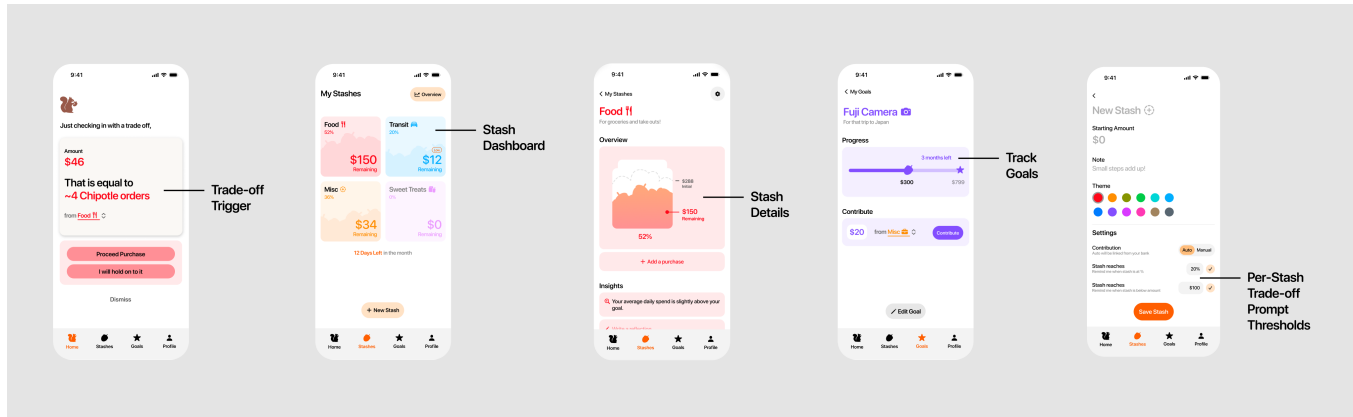


Figure 2: An overview of Nutworth’s core features. The Nutworth interface uses a squirrel companion and value-based “stashes” to contextualize spending behaviors, making abstract trade-offs between types of spending tangible.

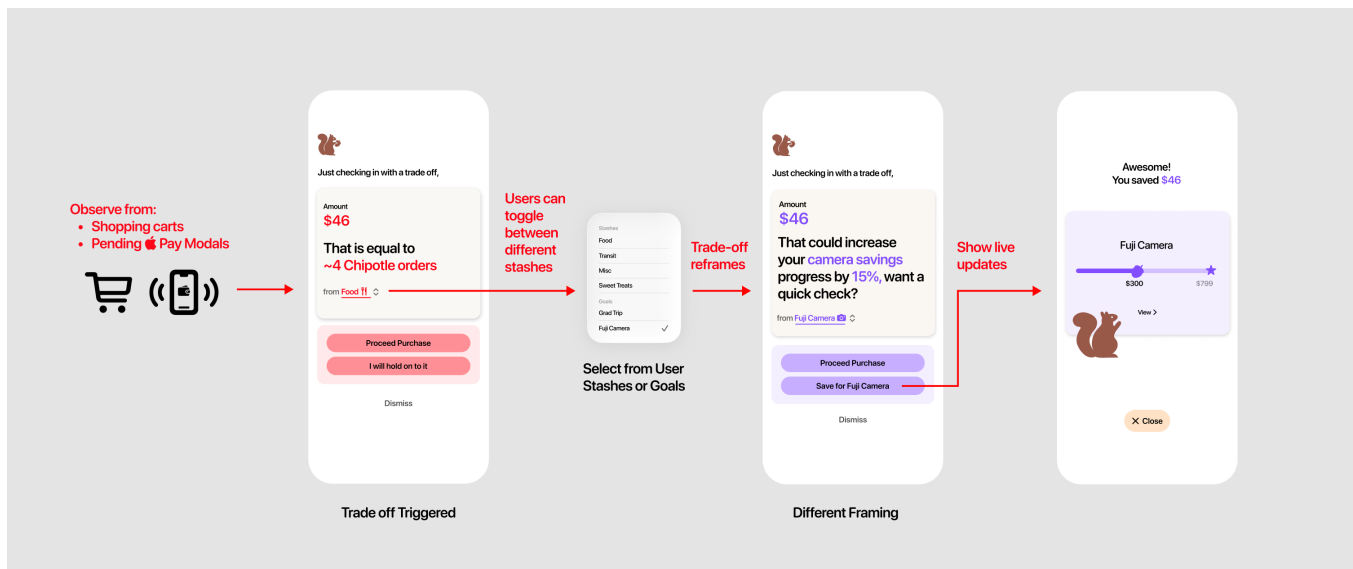


Figure 3: Trade-off trigger flow demonstrating how contextual signals surface reflective prompts and value-based purchase alternatives.

carefully designed to foster internal motivation rather than dependence on external reward. If the user chooses to proceed with the purchase, the system does not punish or shame them. Instead, they are taken to a brief reflection screen that invites optional journaling: “Notes, thoughts, reflections...” This aligns with dual-process theory by encouraging deeper system-led processing in an environment free from pressure. It also respects that not all decisions require reflection; journaling remains available but is always optional.

Rather than pushing users toward predefined financial ideals, Nutworth helps them navigate their own priorities, bringing clarity to the relationship between daily decisions and long-term intentions. It demonstrates that persuasion can be gentle, playful, and collaborative and that financial reflection can feel less like self-surveillance and more like self-discovery. When designed with

empathy and respect for human complexity, financial tools can support users not only in managing money but also in becoming the people they want to be.

4 Reflection

Through conducting three rounds of iterative design, we identified several insights into how young adults engage with emotionally sensitive financial tools. First, psychological safety emerged as a foundational requirement. In early testing, participants disengaged when visuals or language implied surveillance, restriction, or financial punishment, highlighting sensitivity to being monitored or judged. Conversely, they reacted positively when tone, metaphors, and interactions felt warm and autonomy-supportive. This shift informed our move toward more friendly representations

and optional reflective prompts, suggesting that persuasive financial systems must prioritize emotional safety before attempting behavior change. Second, users required a clear conceptual separation between short-term spending and long-term savings. Repeated confusion around stashes and goals disrupted mental models until we refined naming, visual metaphors, and structural explanations, underscoring the importance of coherent conceptual scaffolding in value-based systems. Third, the timing and tone of the intervention proved critical. Participants were more receptive to prompts that felt timely, contextual, and lightweight rather than mandatory, indicating that reflection is more effective when it preserves agency and reduces cognitive burden. Finally, personalization through stash naming, custom colors, and configuration of contribution styles increased emotional investment, but excessive personalization may risk reinforcing existing spending biases or limiting reflection to prior preferences. Framing Nutworth as a supportive companion rather than an authoritative monitor remains significant, alongside privacy-conscious design to preserve agency and psychological safety.

4.1 Open Research Questions for Future Study

Although early findings suggest promise, understanding the long-term impact of interactive systems is important. Reflecting on our iterative progress raises several open questions.

- (1) How can Nutworth sustain long-term changes in spending behavior? Does framing expenses as value-based trade-offs measurably reduce impulsive purchasing or increase consistency over time? When spending and restraint are presented as equally acceptable, does the trade-off prompt encourage genuine reflection, or merely reduce the guilt of proceeding with a purchase?
- (2) How does Nutworth balance being helpful versus intrusive, especially for frequent spenders? How can Nutworth personalize interventions based on different motivational styles, and what are the potential longitudinal effects on financial well-being and emotional outcomes, such as regret, guilt, and agency?
- (3) How can Generative AI be seamlessly integrated to support personalized financial guidance? There is a lack of attention given to financial systems integrating Generative AI [15]. How can Nutworth open the opportunity space for Generative AI for users' spending behavior?

5 Conclusion and Summary

In this paper, we designed Nutworth through an iterative, theory-informed design process to address common emotional and cognitive barriers in student financial decision-making. Through three rounds of user testing, we refined the onboarding, visual metaphors, and trade-off interactions to improve conceptual clarity, reduce guilt, and support reflective spending. Nutworth reframes budgeting from a restrictive system into a supportive companion that encourages financial choices based on personal values.

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